

Derek C. Welty

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Summary

Senior hardware design engineer with 13+ years shipping consumer electronics across networking, display, gaming, and automotive platforms. Owns electrical architecture from schematic and PCB layout through bring-up and validation, integrating wireless, power, audio, camera, and sensor subsystems into compact product designs. Drives component selection and trade-offs across power, cost, performance, and schedule, and partners with ODMs and contract manufacturers from concept to mass production.

Experience

Senior Hardware Design Engineer, Devices

San Francisco, CA

HP IQ, AI INNOVATION LAB

June 2026 – Present

- Driving schematic design, PCB layout, bring-up, and validation for new device hardware
- Collaborating across hardware, firmware, mechanical, and security teams to take new devices from concept to production

Systems Engineering Lead

San Francisco, CA

EERO LLC

June 2025 – June 2026

- Owned system architecture for multi-gig Ethernet switch platforms with PoE, defining power budgets, thermal strategy, and grounding/isolation across multiple SKUs
- Drove thermal architecture and fan control policy, balancing acoustic targets against component derating and reliability constraints
- Directed NPI engagement with contract manufacturer in Taiwan, establishing build-readiness criteria, test strategy, and cross-site engineering workflows
- Led design reviews and mentored engineers across concurrent programs, driving PHY/ASIC integration, compliance decisions, and system-level debug

Senior Display Electrical Engineer

Cupertino, CA

APPLE INC.

Nov 2023 – Feb 2025

- Implemented electrical designs for displays through BOM component selection, schematic creation, directing layout, simulation, and bring-up/debugging circuits in the lab
- Collaborated with a large number of cross-functional groups including PD, DFM, RF, SW, HW Test, System, and EPM to bring a product from concept to mass production
- Defined test criteria and executed test plans on functional prototypes to demonstrate design robustness
- Led external vendors through NPI including build readiness, quality improvements, and root cause FACA analysis

Senior Electrical Engineer

Redmond, WA

MICROSOFT CORPORATION

Jan 2022 – Oct 2023

- Led and managed high volume product design decisions, team members, vendors, and partners
- Motherboard PCBA lead driving sound technical solutions through large cross-disciplinary teams including Power/Thermal, Power Delivery, Silicon, HSIO, LSIO, RF, Wireless, Desense, AV, ME, SW/FW, DFM, Memory, Storage, and HW test
- Led the successful integration of a critical piece of custom silicon from tapeout to full system validation
- Responsible for XBOX console schematic integration, board layout design, and BOM management
- Drove system definition, validation, and bug triage to ensure quality, cost, schedule targets are all met

Senior Design Verification Engineer

Redmond, WA

MICROSOFT CORPORATION

Sept 2019 – Jan 2022

- Owned targeted verification of XBOX systems and subsystems including SoC power management firmware
- Designed, integrated, and maintained automated test systems (Python, PowerShell, C++)
- Performed hardware prototype bring-up, debugging, failure analysis, and telemetry forensics
- Formulated statistical techniques to analyze large data sets and design experiments
- Triage and diagnosed countless high priority issues leading to increased customer experience and factory yield

Senior Design Engineer

Olathe, KS

GARMIN INTERNATIONAL

Mar 2017 – Sept 2019

- Conceptualized and delivered automotive mass production consumer infotainment electronics
- Designed schematics and developed complex rigid/flex printed circuit boards with analog, digital, high-speed, audio, display, camera, wireless, DC-DC power, capacitive multi-touch and RF circuitry
- Project Engineer leading system design and coordinating interdisciplinary collaboration across teams
- Researched components for ideal part selection and BOM management while balancing performance and cost
- Directed and collaborated with factory team in Taiwan to design for testing and manufacturability
- Tested and debugged field hardware/software issues for technical root cause and corrective actions

Experience (cont.)

Electrical Engineer I/II/III
DOE NATIONAL SECURITY CAMPUS, MANAGED BY HONEYWELL

Kansas City, MO
Jan 2013 – Mar 2017

- Lead test engineer architecting/implementing hardware/software solutions for scalable test infrastructure
- Defined test plans and electrical verification methodologies for board level and top-level systems
- Integrated FPGA and CPLD designs into motherboard designs for embedded control and testing
- Interfaced with government customers and supported earned value management project planning

Education

MS in Electrical Engineering
MISSOURI UNIVERSITY OF SCIENCE & TECHNOLOGY

GPA: 4.00/4.00
Aug 2013 – May 2018

Digital Network Design, Wireless Networks, Resilient Networks, Network Performance Analysis and Modeling,
Stochastic Signal Analysis, Power Electronics, Advanced Electromagnetics, Microwave Engineering, Analog CMOS Design

BS in Electrical Engineering, Minor in Mathematics
MISSOURI UNIVERSITY OF SCIENCE & TECHNOLOGY

GPA: 3.93/4.00
Aug 2008 – Dec 2012

IEEE, Eta Kappa Nu, Missouri Miner Photo Editor, Student Union Board Promotions Director

Skills

Protocols	Ethernet (2.5G/10G), PoE (802.3af/at/bt), USB, DDR, I2C, SPI, MDIO, PCIe, HDMI, MIPI UART, JTAG, CAN, SFP/SFP28, GPS
HW Design	Cadence Allegro, OrCAD, Mentor Graphics, CADSTAR, SPICE, Schematic capture, PCB layout SI/PI, EMI/EMC, Thermal Architecture, DFM/DFT, Surge/ESD/HiPOT, PoE System Design, Grounding/Isolation
Programming	Python, PowerShell, MATLAB, Git, TestStand, VISA, AI-assisted engineering, MCP, LLM workflows
Lab Equipment	Oscilloscopes, multimeters, spectrum/network/logic analyzers, power supplies, PXI, DAQ
Extra Training	Advanced GNSS/GPS, Radar Systems Certificate, MIL-STD 1553, Six Sigma Green Belt
Platforms	Jira, Confluence, Arena PLM, Linux, Google Workspace, Azure DevOps

Awards & Patents

2016	Outstanding Engineer Award , Honeywell Aerospace
2019	US10447297B1 , Electronic Device and Method for Compressing Sampled Data